Flight Delay Intelligence

Dashboard screenshots

2015 US flights · T-60, day-ahead and post-departure prediction · weather impact · base-paper comparison

2 Overview

3 T-60 · Performance

4 T-60 · Weather and drivers

5 T-60 · Live predictor (a real Nov–Dec flight)

6 T-60 · Against the base paper

7 Day-ahead · Performance

8 Day-ahead · Weather and features

9 Day-ahead · Against the base paper

10 Post-departure · Performance

11 Post-departure · Against the base paper

12 Metric evaluation · Scenarios

13 Metric evaluation · Models

14 Metric evaluation · Curves

15 Metric evaluation · Against the base paper

16 Metric evaluation · Tuning

17 Explore · Time and airlines

18 Explore · Weather and causes

19 Explore · Delay propagation

20 Explore · Airports and cancellations

21 About and limits

Light theme, 1440 px wide. Captured from the local portal at http://localhost:8765

FLIGHT DELAY AND AIRPORT OPERATIONS INTELLIGENCE

Why flights are late, and how early we can tell

We built a warehouse from 5.8 million 2015 US flights and NOAA weather to measure how much weather really changes delays and cancellations, and to predict whether a flight will arrive 15 or more minutes late, one hour before departure and the day before.

Flights analysed	Arrive on time	Delayed in severe weather	Cancelled in severe weather
5.82M	81.4%	40.4%	9.0%
January–December 2015	under 15 minutes late	against 17.0% when clear	against 1.1% when clear

Three prediction moments

THE OPERATIONAL FORECAST

T-60 · one hour before departure

0.645

best PR-AUC (Ensemble) · 3.6× chance

Open the results →

THE PLANNING FORECAST

Day-ahead · the day before

0.360

best PR-AUC (Ensemble) · 2.0× chance

Open the results →

THE COMPARISON SCENARIO

Post-departure · after take-off

0.880

best PR-AUC (XGBoost (tuned)) · 4.9× chance

Open the results →

What we found

- **Weather matters.** Severe weather at the origin raises the delayed-flight rate from 17.0% to 40.4% and cancellations from 1.1% to 9.0%.
- **It helps prediction most when little else is known.** Weather features add +0.100 PR-AUC day-ahead and +0.043 at T-60.
- **A late aircraft carries its delay forward.** When the inbound aircraft is 30–59 minutes late, 83.2% of the next departures leave 15+ minutes late.
- **Knowing the aircraft is the biggest lever.** Best PR-AUC rises from 0.360 day-ahead to 0.645 at T-60.
- **Against the base paper.** At our F1 threshold we beat its accuracy in every scenario (up to +16.1 points) but not its recall; see the scorecard.

Where to look next

Try the live predictor →
Pick a flight, or load a real one and see what happened.

Overall metric evaluation →
Every model and scenario: curves, confidence intervals, tuning and the paper scorecard.

Explore the data →
Time patterns, weather, propagation, airports and cancellations.

How it was built

Warehouse	Analysis	Prediction	Explainability	Patterns	This portal
Star schema of flights, airlines, airports, routes and weather in DuckDB.	OLAP, weather impact, root causes and delay propagation.	11 models, a neural network and an ensemble, tuned and tested on later months.	SHAP, ablations and subgroup checks.	Airport clusters and cancellation rules.	Plain JavaScript and d3, with a Python API serving the saved model.

PREDICTION MOMENT · T-60

One hour before departure

The operational forecast: sixty minutes before a flight is scheduled to leave, will it arrive 15 or more minutes late? The model knows the schedule, airline, airports, the weather and what the inbound aircraft is doing.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✓ Inbound aircraft status and delay · ✗ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.645	3.6×	0.824	+0.043	87.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+16.1 points accuracy · −24.2 recall

Performance · Weather and drivers · Live predictor · Against the base paper

Model leaderboard

Test flights, November–December. Bars use a 0–1 scale, so gaps are not exaggerated.

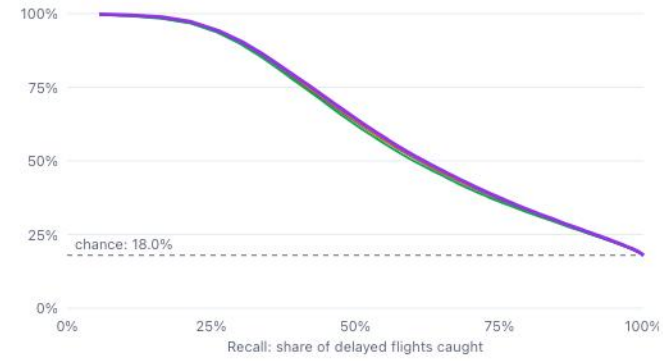
PR-AUC · ROC-AUC · F1 · F2 · Recall · Precision · Accuracy

Ensemble (LightGBM + XGBoost + MLP)	0.645
XGBoost (native categoricals, tuned)	0.641
XGBoost (tuned, ordinal codes)	0.639
LightGBM (main)	0.639
Neural Network (MLP)	0.634
Random Forest	0.619
Decision Tree (CART)	0.604
Logistic Regression	0.550
KNN	0.478
Naive Bayes (Categorical)	0.459
Naive Bayes (Gaussian)	0.362

Primary metric: how well flights are ranked by risk, robust to the 18% delay rate.

Precision against recall

Each line is a model; points are alert levels. Higher and further right is better.



Flag the riskiest flights

Drag to choose how many flights get an alert.

Ensemble

Alert on the riskiest 18% of flights

Precision	56.2%	of alerted flights really are delayed
Recall	56.4%	of all delayed flights get an alert
F2	0.563	recall counts double
Accuracy	84.3%	overall; a poor guide here
Lift	3.1×	versus alerting at random

Nearest computed point: top 18% of Ensemble scores.

All metrics

Threshold tuned for F1 on September and applied unchanged to the test flights. Click a heading to sort.

Download CSV

Model	PR-AUC ▾	ROC-AUC	Accuracy	Precision	Recall	F1	F2
Ensemble (LightGBM + XGBoost + MLP)	**0.645**	**0.824**	**87.0%**	**72.4%**	**44.3%**	**0.549**	**0.480**
XGBoost (native categoricals, tuned)	0.641	0.821	85.2%	60.0%	52.7%	0.561	0.541
XGBoost (tuned, ordinal codes)	0.639	0.819	85.1%	59.4%	53.0%	0.560	0.542
LightGBM (main)	0.639	0.820	84.8%	58.4%	53.8%	0.560	0.547
Neural Network (MLP)	0.634	0.817	85.9%	64.1%	48.7%	0.554	0.512
Random Forest	0.619	0.813	84.9%	59.3%	51.2%	0.550	0.527
Decision Tree (CART)	0.604	0.793	85.7%	64.7%	45.3%	0.533	0.482
Logistic Regression	0.550	0.781	82.8%	52.1%	49.4%	0.507	0.499
KNN	0.478	0.740	82.4%	51.2%	41.7%	0.460	0.433
Naive Bayes (Categorical)	0.459	0.732	82.4%	51.2%	36.9%	0.429	0.391
Naive Bayes (Gaussian)	0.362	0.720	71.8%	34.0%	60.7%	0.436	0.524

What each extra piece of information buys

Best PR-AUC of any model in each scenario; the dashed line is the 18% chance level.



PREDICTION MOMENT · T-60

One hour before departure

The operational forecast: sixty minutes before a flight is scheduled to leave, will it arrive 15 or more minutes late? The model knows the schedule, airline, airports, the weather and what the inbound aircraft is doing.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✓ Inbound aircraft status and delay · ✗ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.645	3.6×	0.824	+0.043	87.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+16.1 points accuracy · −24.2 recall

Performance | **Weather and drivers** | Live predictor | Against the base paper

What weather adds

PR-AUC with and without the weather features, tuned models.

XGBoost (PSO-tuned)

PR-AUC 0.599 without weather → 0.641 with it (+0.042, 95% interval 0.041 to 0.043)

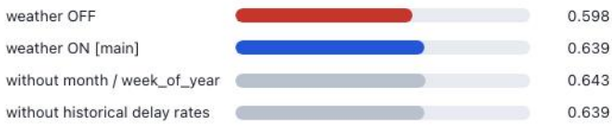
LightGBM (PSO-tuned)

PR-AUC 0.599 without weather → 0.642 with it (+0.043, 95% interval 0.043 to 0.044)

> Intervals are 95% paired bootstrap ranges and exclude zero. Weather is observed same-day weather, an upper bound on what a real forecast would give.

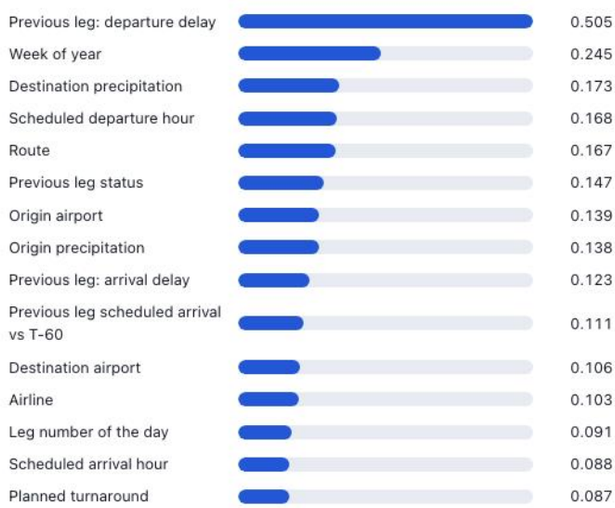
Feature ablation

PR-AUC of LightGBM as feature groups are removed or added.



What drives the T-60 prediction

Mean absolute SHAP value, top features.



Share of model gain by feature group

LightGBM.



Where the model is strongest and weakest

LightGBM on test flights, by subgroup. Download CSV

Subgroup	Flights	Delayed	PR-AUC ▾	ROC-AUC
Predecessor 'not_departed' at T-60	15,679	95.4%	0.991	0.871
Origin snow > 0	30,259	40.5%	0.748	0.775
All test flights	932,084	18.0%	0.639	0.820
Rare routes (<100 training flights)	10,822	19.3%	0.629	0.809
Origin weather missing (SJU, STT, ... NaN weather)	6,927	15.1%	0.587	0.808
No predecessor (first leg / broken chain)	235,496	14.9%	0.380	0.729
Predecessor 'landed' at T-60	280,440	9.4%	0.236	0.716

PREDICTION MOMENT · T-60

One hour before departure

The operational forecast: sixty minutes before a flight is scheduled to leave, will it arrive 15 or more minutes late? The model knows the schedule, airline, airports, the weather and what the inbound aircraft is doing.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✓ Inbound aircraft status and delay · ✗ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.645	3.6×	0.824	+0.043	87.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+16.1 points accuracy · −24.2 recall

Performance · Weather and drivers · **Live predictor** · Against the base paper

Predict a flight Runs the saved T-60 model (LightGBM, 417 trees). Choose a flight and what is known about the inbound aircraft, or load a real November–December flight and compare with what happened.

Origin airport	Destination airport	Airline	Month	Day of week	Scheduled departure
SFO	FLL	UA · United Air Lines In	Dec	Saturday	22:00

Origin weather	Destination weather	Inbound aircraft at T-60	Inbound left its gate (min late)	Planned turnaround (min)	Leg of the day
Clear	Clear	Inbound aircraft is in th	-9	60	4

Predict · Load a real Nov–Dec flight

Model on the flight's exact recorded data

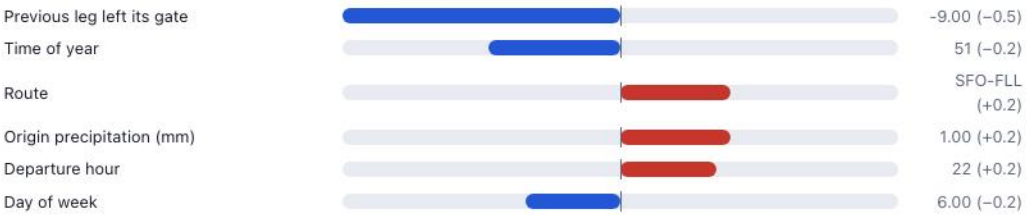
39%

Likely on time

Alert line at 57% (tuned on September). This flight is 2.2× the 18% delay rate of the test period.

What pushed the estimate

Right raises the risk, left lowers it (SHAP, log-odds).



What actually happened on 2015-12-19: it arrived 29 min early, so it was on time. The model was **correct** at the alert line. One flight says little about accuracy; see the leaderboard.

Rebuilt from the form fields (weather as a typical class value): 31%.

PREDICTION MOMENT · T-60

One hour before departure

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Performance · Weather and drivers · Live predictor · **Against the base paper**

Our F1-tuned threshold | Recall matched to paper — Green marks where we beat the paper's XGBoost; red where we do not.

> Our models use their own threshold, tuned for F1 on September. It is not adjusted to the paper's operating point: the paper's undersampling pushes recall up and accuracy down, so expect higher accuracy and lower recall here.

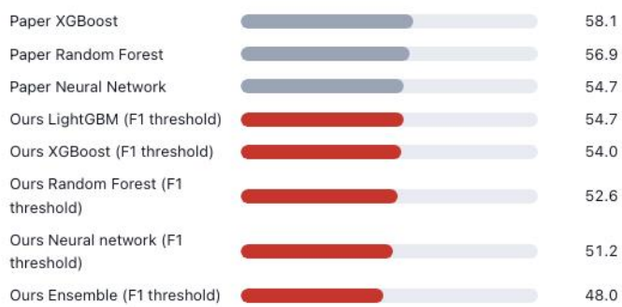
Accuracy %



Recall % of delayed flights caught



F2 score recall counts double



Precision the paper does not report it



Side by side

Differences are percentage points against the paper's XGBoost. Blank cells are not reported by the paper. Download CSV

System	Accuracy	Precision	Recall	F2	PR-AUC	ROC-AUC	Δ accuracy	Δ recall	Δ F2
Paper XGBoost	70.9		68.5	58.1					
Paper Random Forest	70.1		67.2	56.9					
Paper Neural Network	66.6		66.5	54.7					
Ours LightGBM (F1 threshold)	84.8	58.4	53.8	54.7	0.639	0.820	+13.9	−14.7	−3.4
Ours XGBoost (F1 threshold)	85.2	60.0	52.7	54.0	0.641	0.821	+14.3	−15.8	−4.1
Ours Random Forest (F1 threshold)	84.9	59.3	51.2	52.6	0.619	0.813	+14.0	−17.3	−5.5
Ours Neural network (F1 threshold)	85.9	64.1	48.7	51.2	0.634	0.817	+15.0	−19.8	−6.9
Ours Ensemble (F1 threshold)	87.0	72.4	44.3	48.0	0.645	0.824	+16.1	−24.2	−10.1

PREDICTION MOMENT · DAY-AHEAD

The day before

The planning forecast: using only what is known in advance, which flights are at risk of arriving 15 or more minutes late? Nothing is known about the aircraft's earlier legs, so this is the hardest of the three tasks.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✗ Inbound aircraft status and delay · ✗ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.360	2.0×	0.703	+0.100	75.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+5.5 points accuracy · −23.7 recall

Performance | Weather and features | Against the base paper

Model leaderboard

Test flights, November–December. Bars use a 0–1 scale, so gaps are not exaggerated.

PR-AUC | ROC-AUC | F1 | F2 | Recall | Precision | Accuracy

Ensemble (LightGBM + XGBoost + MLP)	0.360
XGBoost (native categoricals, tuned)	0.356
LightGBM (main)	0.354
Random Forest	0.342
XGBoost (tuned, ordinal codes)	0.341
Neural Network (MLP)	0.337
Logistic Regression	0.320
Naive Bayes (Gaussian)	0.288
KNN	0.285
Decision Tree (CART)	0.285
Naive Bayes (Categorical)	0.271

> Primary metric: how well flights are ranked by risk, robust to the 18% delay rate.

Precision against recall

Each line is a model; points are alert levels. Higher and further right is better.



Flag the riskiest flights

Drag to choose how many flights get an alert.

Ensemble

Alert on the riskiest 18% of flights

Precision	37.4%	of alerted flights really are delayed
Recall	37.5%	of all delayed flights get an alert
F2	0.375	recall counts double
Accuracy	77.5%	overall; a poor guide here
Lift	2.1×	versus alerting at random

> Nearest computed point: top 18% of Ensemble scores.

All metrics

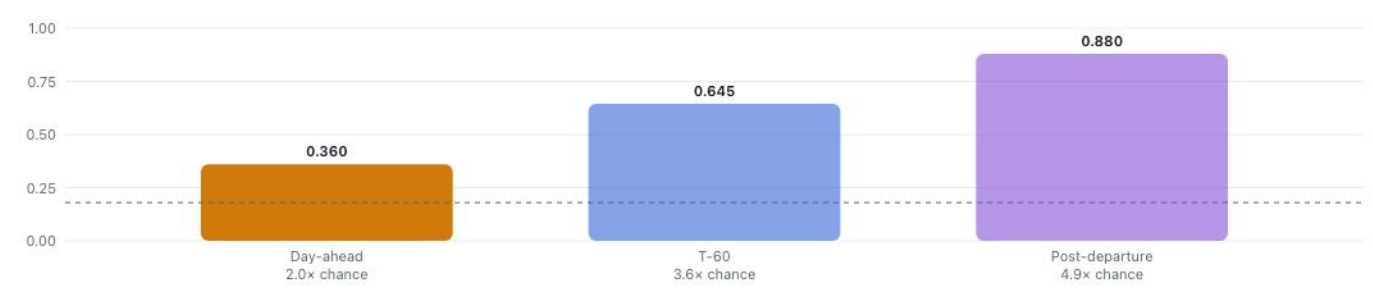
Threshold tuned for F1 on September and applied unchanged to the test flights. Click a heading to sort.

Download CSV

Model	PR-AUC ▾	ROC-AUC	Accuracy	Precision	Recall	F1	F2
Ensemble (LightGBM + XGBoost + MLP)	**0.360**	**0.703**	**75.0%**	**34.7%**	**44.5%**	**0.390**	**0.421**
XGBoost (native categoricals, tuned)	0.356	0.700	69.8%	30.9%	55.3%	0.397	0.478
LightGBM (main)	0.354	0.699	70.3%	31.2%	54.1%	0.396	0.472
Random Forest	0.342	0.693	68.3%	29.8%	56.7%	0.391	0.480
XGBoost (tuned, ordinal codes)	0.341	0.687	65.8%	28.5%	59.7%	0.385	0.489
Neural Network (MLP)	0.337	0.689	73.4%	32.5%	44.8%	0.377	0.416
Logistic Regression	0.320	0.670	69.8%	29.7%	49.6%	0.371	0.437
Naive Bayes (Gaussian)	0.288	0.653	49.5%	22.8%	75.8%	0.350	0.517
KNN	0.285	0.623	71.8%	28.5%	37.7%	0.325	0.354
Decision Tree (CART)	0.285	0.637	73.1%	29.7%	36.6%	0.328	0.350
Naive Bayes (Categorical)	0.271	0.628	74.0%	29.1%	30.9%	0.300	0.305

What each extra piece of information buys

Best PR-AUC of any model in each scenario; the dashed line is the 18% chance level.



PREDICTION MOMENT · DAY-AHEAD

The day before

The planning forecast: using only what is known in advance, which flights are at risk of arriving 15 or more minutes late? Nothing is known about the aircraft's earlier legs, so this is the hardest of the three tasks.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✗ Inbound aircraft status and delay · ✗ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.360	2.0×	0.703	+0.100	75.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+5.5 points accuracy · −23.7 recall

Performance | **Weather and features** | Against the base paper

What weather adds

PR-AUC with and without the weather features, tuned models.

XGBoost (PSO-tuned)

PR-AUC 0.247 without weather → 0.344 with it (+0.097, 95% interval 0.095 to 0.099)

LightGBM (PSO-tuned)

PR-AUC 0.244 without weather → 0.347 with it (+0.103, 95% interval 0.101 to 0.105)

> Intervals are 95% paired bootstrap ranges and exclude zero. Weather is observed same-day weather, an upper bound on what a real forecast would give.

Feature ablation

PR-AUC of LightGBM as feature groups are removed or added.



Reading the ablation: weather lifts PR-AUC from 0.248 to 0.354, and knowing the planned turnaround structure from the schedule adds a further 0.064. Feature-level SHAP charts were computed for the T-60 model; open the T-60 page for them.

PREDICTION MOMENT · DAY-AHEAD

The day before

The planning forecast: using only what is known in advance, which flights are at risk of arriving 15 or more minutes late? Nothing is known about the aircraft's earlier legs, so this is the hardest of the three tasks.

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Best PR-AUC	Lift over chance	Best ROC-AUC	Weather adds	Best accuracy vs paper XGBoost
0.360	2.0×	0.703	+0.100	75.0%
Ensemble (LightGBM + XGBoost + MLP)	chance is 18.0% delayed	Ensemble (LightGBM + XGBoost + MLP)	PR-AUC, tuned models	+5.5 points accuracy · −23.7 recall

Performance · Weather and features · Against the base paper

Our F1-tuned threshold | Recall matched to paper

Green marks where we beat the paper's XGBoost; red where we do not.

> Our models use their own threshold, tuned for F1 on September. It is not adjusted to the paper's operating point: the paper's undersampling pushes recall up and accuracy down, so expect higher accuracy and lower recall here.

Accuracy %

Paper XGBoost	69.5
Paper Random Forest	68.7
Paper Neural Network	66.1
Ours LightGBM (F1 threshold)	70.3
Ours XGBoost (F1 threshold)	69.8
Ours Random Forest (F1 threshold)	68.3
Ours Neural network (F1 threshold)	73.4
Ours Ensemble (F1 threshold)	75.0

Recall % of delayed flights caught

Paper XGBoost	68.2
Paper Random Forest	67.2
Paper Neural Network	65.8
Ours LightGBM (F1 threshold)	54.1
Ours XGBoost (F1 threshold)	55.3
Ours Random Forest (F1 threshold)	56.6
Ours Neural network (F1 threshold)	44.8
Ours Ensemble (F1 threshold)	44.5

F2 score recall counts double

Paper XGBoost	57.2
Paper Random Forest	56.2
Paper Neural Network	54.0
Ours LightGBM (F1 threshold)	47.2
Ours XGBoost (F1 threshold)	47.8
Ours Random Forest (F1 threshold)	48.0
Ours Neural network (F1 threshold)	41.6
Ours Ensemble (F1 threshold)	42.1

Precision the paper does not report it

Ours LightGBM (F1 threshold)	31.2
Ours XGBoost (F1 threshold)	30.9
Ours Random Forest (F1 threshold)	29.8
Ours Neural network (F1 threshold)	32.5
Ours Ensemble (F1 threshold)	34.7

Side by side Differences are percentage points against the paper's XGBoost. Blank cells are not reported by the paper. Download CSV

System	Accuracy	Precision	Recall	F2	PR-AUC	ROC-AUC	Δ accuracy	Δ recall	Δ F2
Paper XGBoost	69.5		68.2	57.2					
Paper Random Forest	68.7		67.2	56.2					
Paper Neural Network	66.1		65.8	54.0					
Ours LightGBM (F1 threshold)	70.3	31.2	54.1	47.2	0.354	0.699	+0.8	−14.1	−10.0
Ours XGBoost (F1 threshold)	69.8	30.9	55.3	47.8	0.356	0.700	+0.3	−12.9	−9.4
Ours Random Forest (F1 threshold)	68.3	29.8	56.6	48.0	0.342	0.693	−1.3	−11.6	−9.2
Ours Neural network (F1 threshold)	73.4	32.5	44.8	41.6	0.337	0.689	+3.9	−23.4	−15.6
Ours Ensemble (F1 threshold)	75.0	34.7	44.5	42.1	0.360	0.703	+5.5	−23.7	−15.1

> The paper is ahead on this task. Its random split and 30-airport subset make day-ahead prediction easier than our time-based test over all airports.

COMPARISON SCENARIO · POST-DEPARTURE

After take-off

The flight's own departure delay is added. It is only known after take-off, so this is not an operational forecast. It exists to compare with the base paper's third feature set and to show how much that one number tells you.

✓ Schedule and calendar · ✓ Airline, airport, route · ✓ Observed weather · ✓ Inbound aircraft status and delay · ✓ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Departure delay adds	Best accuracy vs paper XGBoost
0.880	4.9×	0.944	+0.235	93.1%
XGBoost (tuned)	chance is 18.0% delayed	XGBoost (tuned)	PR-AUC over the T-60 best	+3.2 points accuracy · −7.7 recall

Performance | Against the base paper

Model leaderboard

Test flights, November–December. Bars use a 0–1 scale, so gaps are not exaggerated.

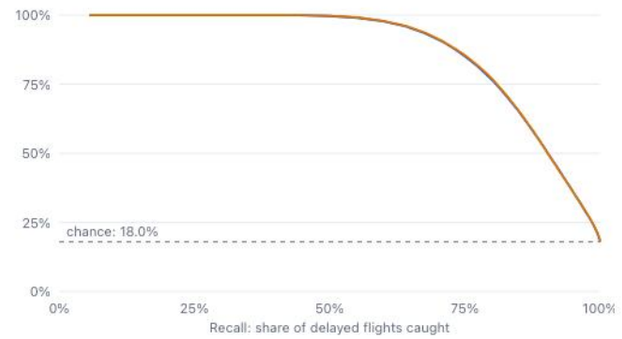
PR-AUC | ROC-AUC | F1 | F2 | Recall | Precision | Accuracy

XGBoost (tuned)	0.880
LightGBM	0.879

Primary metric: how well flights are ranked by risk, robust to the 18% delay rate.

Precision against recall

Each line is a model; points are alert levels. Higher and further right is better.



Flag the riskiest flights

Drag to choose how many flights get an alert.

LightGBM

Alert on the riskiest 18% of flights

Precision	78.7%	of alerted flights really are delayed
Recall	78.9%	of all delayed flights get an alert
F2	0.788	recall counts double
Accuracy	92.4%	overall; a poor guide here
Lift	4.4×	versus alerting at random

Nearest computed point: top 18% of LightGBM scores.

All metrics

Threshold tuned for F1 on September and applied unchanged to the test flights. Click a heading to sort.

Download CSV

Model	PR-AUC ▾	ROC-AUC	Accuracy	Precision	Recall	F1	F2
XGBoost (tuned)	**0.880**	**0.944**	**93.1%**	**84.4%**	**75.8%**	**0.799**	**0.773**
LightGBM	0.879	0.943	93.0%	83.9%	75.9%	0.797	0.773

What each extra piece of information buys

Best PR-AUC of any model in each scenario; the dashed line is the 18% chance level.



COMPARISON SCENARIO · POST-DEPARTURE

After take-off

The flight's own departure delay is added. It is only known after take-off, so this is not an operational forecast. It exists to compare with the base paper's third feature set and to show how much that one number tells you.

✓ Schedule and calendar ✓ Airline, airport, route ✓ Observed weather ✓ Inbound aircraft status and delay ✓ This flight's own departure delay

Best PR-AUC	Lift over chance	Best ROC-AUC	Departure delay adds	Best accuracy vs paper XGBoost
0.880	4.9×	0.944	+0.235	93.1%
XGBoost (tuned)	chance is 18.0% delayed	XGBoost (tuned)	PR-AUC over the T-60 best	+3.2 points accuracy · −7.7 recall

Performance | Against the base paper

Our F1-tuned threshold | Recall matched to paper

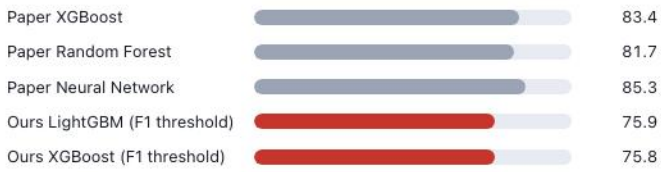
Green marks where we beat the paper's XGBoost; red where we do not.

> Our models use their own threshold, tuned for F1 on September. It is not adjusted to the paper's operating point: the paper's undersampling pushes recall up and accuracy down, so expect higher accuracy and lower recall here.

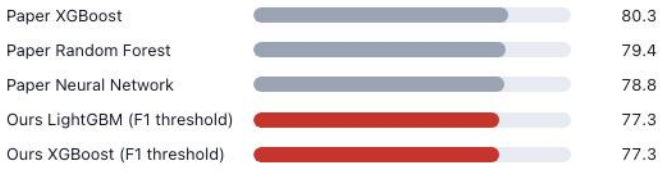
Accuracy %



Recall % of delayed flights caught



F2 score recall counts double



Precision the paper does not report it



Side by side Differences are percentage points against the paper's XGBoost. Blank cells are not reported by the paper. Download CSV

System	Accuracy	Precision	Recall	F2	PR-AUC	ROC-AUC	Δ accuracy	Δ recall	Δ F2
Paper XGBoost	89.9		83.4	80.3					
Paper Random Forest	90.2		81.7	79.4					
Paper Neural Network	86.5		85.3	78.8					
Ours LightGBM (F1 threshold)	93.0	83.9	75.9	77.3	0.879	0.943	+3.1	−7.5	−3.0
Ours XGBoost (F1 threshold)	93.1	84.4	75.8	77.3	0.880	0.944	+3.2	−7.7	−3.0

OVERALL METRIC EVALUATION

Every model, every scenario, in graphs

PR-AUC is the main metric: it stays honest when only 18% of flights are late. Precision, recall, F1 and F2 use the threshold tuned on September; ROC-AUC and the curves do not depend on any threshold.

Scenarios | Models | Curves | Against the base paper | Tuning

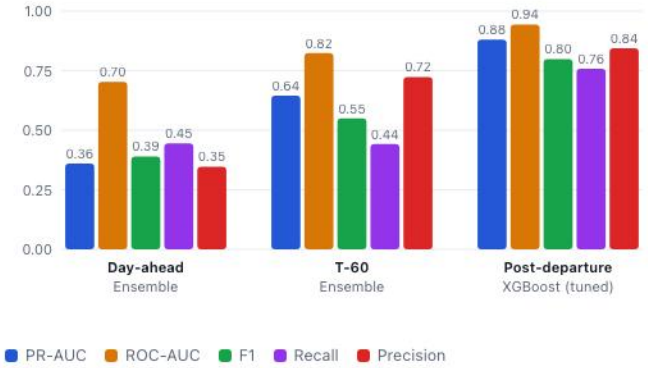
Best PR-AUC in each scenario

More information, better ranking. Dashed line: the 18% chance level.



Best model per scenario, by metric

The best-PR-AUC model in each scenario.



What moves performance

PR-AUC gained from each piece of information (weather bars carry a 95% interval in the tooltip).



OVERALL METRIC EVALUATION

Every model, every scenario, in graphs

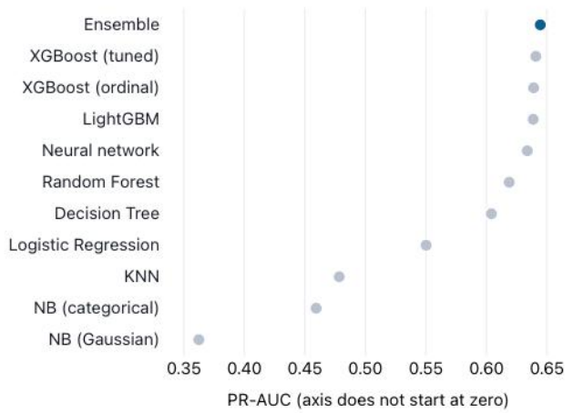
PR-AUC is the main metric: it stays honest when only 18% of flights are late. Precision, recall, F1 and F2 use the threshold tuned on September; ROC-AUC and the curves do not depend on any threshold.

Scenarios | Models | Curves | Against the base paper | Tuning

Day-ahead | T-60 | Post-departure

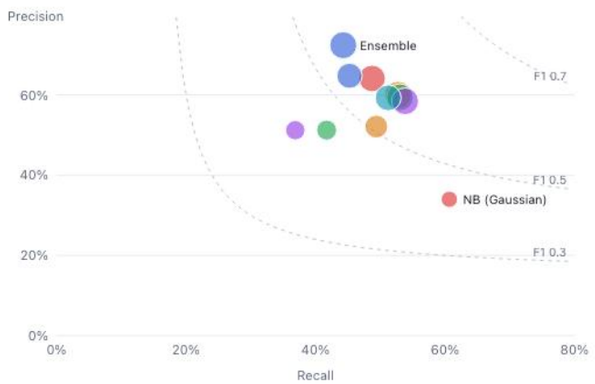
PR-AUC with 95% confidence intervals

Dots are PR-AUC, whiskers the bootstrap interval over test flights. Neural network, ensemble and post-departure models have no interval.



Precision against recall

Each bubble is a model at its F1 threshold; size is PR-AUC. Dashed curves are constant F1.



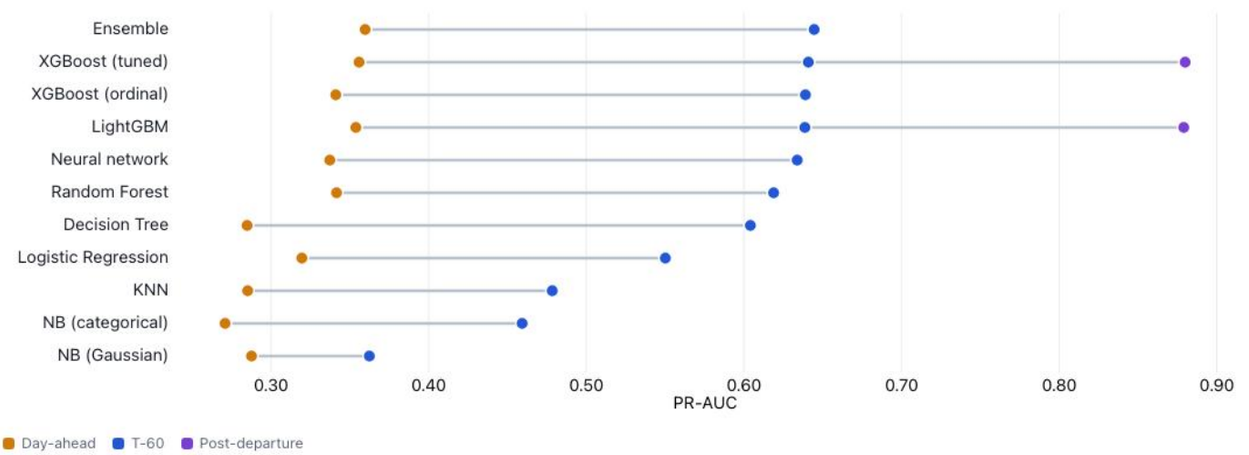
Model × metric heatmap

Darker is better within each column. Click a heading to sort. Download CSV

Model	PR-AUC ▾	ROC-AUC	F1	F2	Recall	Precision	Accuracy
Ensemble	0.645	0.824	0.549	0.480	0.443	0.724	0.870
XGBoost (tuned)	0.641	0.821	0.561	0.541	0.527	0.600	0.852
XGBoost (ordinal)	0.639	0.819	0.560	0.542	0.530	0.594	0.851
LightGBM	0.639	0.820	0.560	0.547	0.538	0.584	0.848
Neural network	0.634	0.817	0.554	0.512	0.487	0.641	0.859
Random Forest	0.619	0.813	0.550	0.527	0.512	0.593	0.849
Decision Tree	0.604	0.793	0.533	0.482	0.453	0.647	0.857
Logistic Regression	0.550	0.781	0.507	0.499	0.494	0.521	0.828
KNN	0.478	0.740	0.460	0.433	0.417	0.512	0.824
NB (categorical)	0.459	0.732	0.429	0.391	0.369	0.512	0.824
NB (Gaussian)	0.362	0.720	0.436	0.524	0.607	0.340	0.718

Does the ranking hold across scenarios?

PR-AUC of each model in every scenario it was run in.



OVERALL METRIC EVALUATION

Every model, every scenario, in graphs

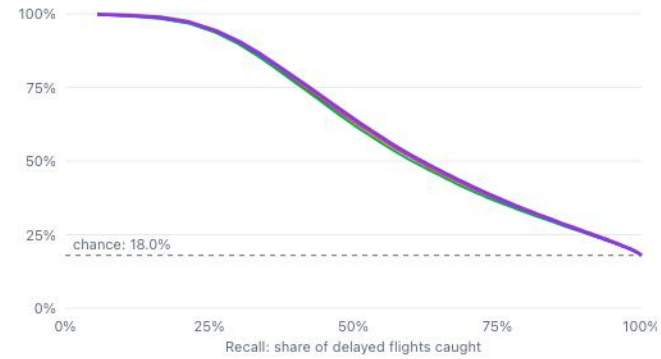
PR-AUC is the main metric: it stays honest when only 18% of flights are late. Precision, recall, F1 and F2 use the threshold tuned on September; ROC-AUC and the curves do not depend on any threshold.

Scenarios | Models | Curves | Against the base paper | Tuning

Day-ahead | T-60 | Post-departure — Sweep model: LightGBM

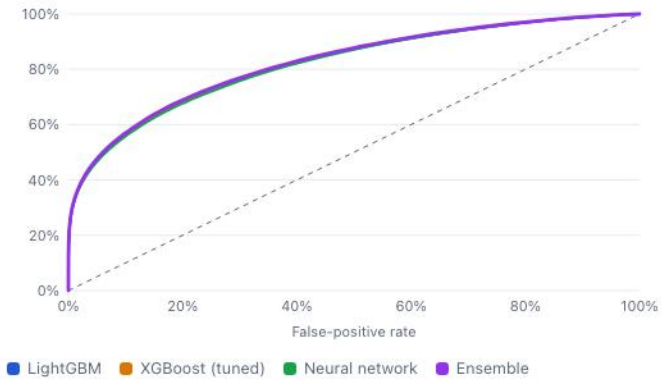
Precision–recall curves

Every threshold at once. Higher and further right is better.



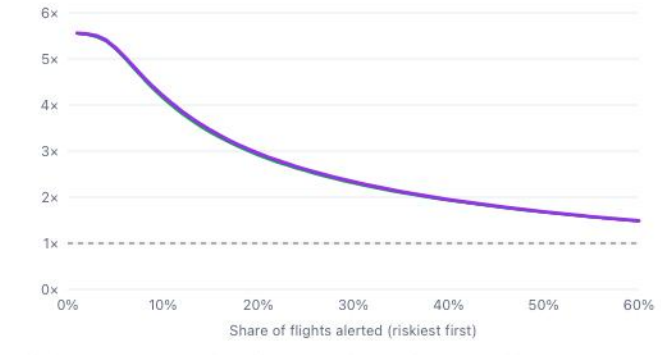
ROC curves

True-positive rate against false-positive rate. The diagonal is chance.



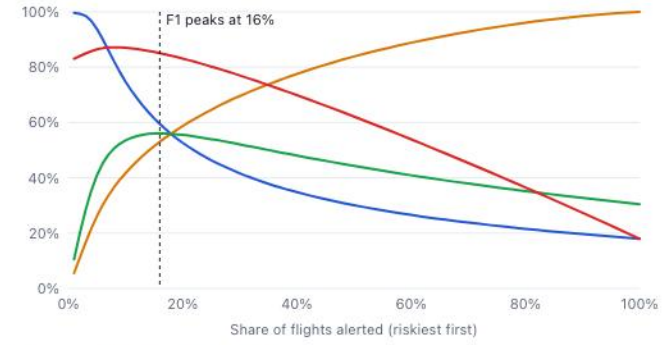
Lift curve

How many times better than random each alert level is.



Threshold sweep

Precision, recall, F1 and accuracy as more flights get an alert. Accuracy peaks when almost nobody is flagged, which is why it is a poor guide.



OVERALL METRIC EVALUATION

Every model, every scenario, in graphs

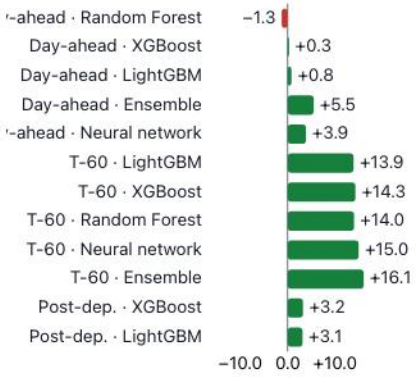
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Scenarios | Models | Curves | **Against the base paper** | Tuning

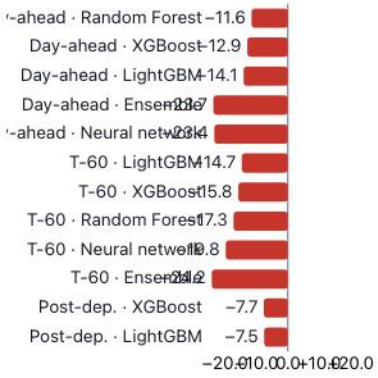
Our F1-tuned threshold | Recall matched to paper

Differences in points against the paper's XGBoost in the same scenario. Green is better than the paper.

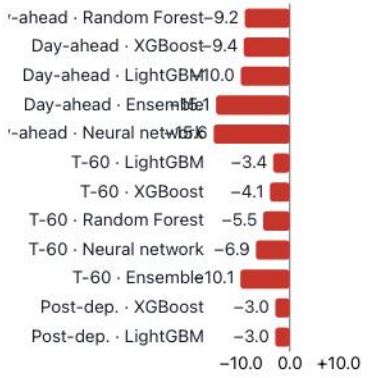
Δ Accuracy points vs paper XGBoost



Δ Recall points vs paper XGBoost



Δ F2 score points vs paper XGBoost



Scorecard

Verdict by F2, the paper's main metric: higher above +0.5 points, level within ±0.5, lower below.

Download CSV

> Of 12 systems: 11 beat the paper on accuracy, 0 on recall, 0 on F2 (0 level). At our own threshold we gain accuracy and give up recall; that is a threshold choice, not weaker ranking.

Scenario ▲	Our system	Paper acc.	Our acc.	Δ acc.	Paper recall	Our recall	Δ recall	Paper F2	Our F2	Δ F2	Verdict
Day-ahead	Ours Random Forest (F1 threshold)	69.5	68.3	−1.3	68.2	56.6	−11.6	57.2	48.0	−9.2	Lower
Day-ahead	Ours XGBoost (F1 threshold)	69.5	69.8	+0.3	68.2	55.3	−12.9	57.2	47.8	−9.4	Lower
Day-ahead	Ours LightGBM (F1 threshold)	69.5	70.3	+0.8	68.2	54.1	−14.1	57.2	47.2	−10.0	Lower
Day-ahead	Ours Ensemble (F1 threshold)	69.5	75.0	+5.5	68.2	44.5	−23.7	57.2	42.1	−15.1	Lower
Day-ahead	Ours Neural network (F1 threshold)	69.5	73.4	+3.9	68.2	44.8	−23.4	57.2	41.6	−15.6	Lower
Post-departure	Ours XGBoost (F1 threshold)	89.9	93.1	+3.2	83.4	75.8	−7.7	80.3	77.3	−3.0	Lower
Post-departure	Ours LightGBM (F1 threshold)	89.9	93.0	+3.1	83.4	75.9	−7.5	80.3	77.3	−3.0	Lower
T-60	Ours LightGBM (F1 threshold)	70.9	84.8	+13.9	68.5	53.8	−14.7	58.1	54.7	−3.4	Lower
T-60	Ours XGBoost (F1 threshold)	70.9	85.2	+14.3	68.5	52.7	−15.8	58.1	54.0	−4.1	Lower

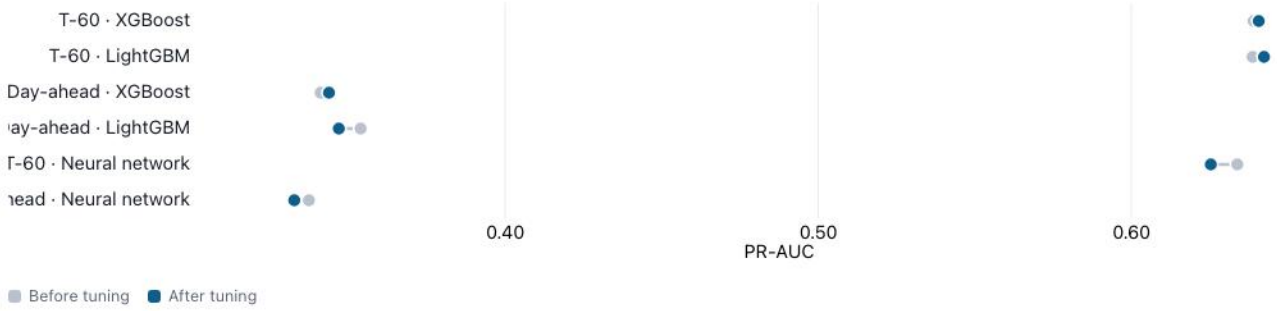
OVERALL METRIC EVALUATION

Every model, every scenario, in graphs

PR-AUC is the main metric: it stays honest when only 18% of flights are late. Precision, recall, F1 and F2 use the threshold tuned on September; ROC-AUC and the curves do not depend on any threshold.

Scenarios Models Curves Against the base paper Tuning

PR-AUC before and after particle swarm tuning Tuning added at most +0.004 and hurt some models: settings chosen on a 600,000-row sample did not carry over to full-data training.



Tuned values Download CSV

Scenario ▲	Model	Tuned values	Before	After	Change
Day-ahead	XGBoost	max_depth 9 · learning_rate 0.05 · min_child_weight 55.7 · colsample_bytree 0.4 · gamma 1.02 · subsample 1	0.341	0.344	+0.0027
Day-ahead	LightGBM	num_leaves 78 · learning_rate 0.0984 · min_child_samples 2000 · feature_fraction 0.767 · lambda_l2 0.815 · bagging_fraction 1	0.354	0.347	−0.0070
Day-ahead	Neural network	width 781 · drop 0.199 · lr 0.00383 · wd 0.000212 · bs 1024	0.337	0.333	−0.0047
T-60	XGBoost	max_depth 9 · learning_rate 0.0546 · min_child_weight 40 · colsample_bytree 0.48 · gamma 0.106 · subsample 0.989	0.639	0.641	+0.0016
T-60	LightGBM	num_leaves 164 · learning_rate 0.05 · min_child_samples 674 · feature_fraction 0.4 · lambda_l2 0.01 · bagging_fraction 1	0.639	0.642	+0.0036
T-60	Neural network	width 676 · drop 0.103 · lr 0.00757 · wd 8.82e-05 · bs 1024	0.634	0.625	−0.0085

EXPLORE THE DATA

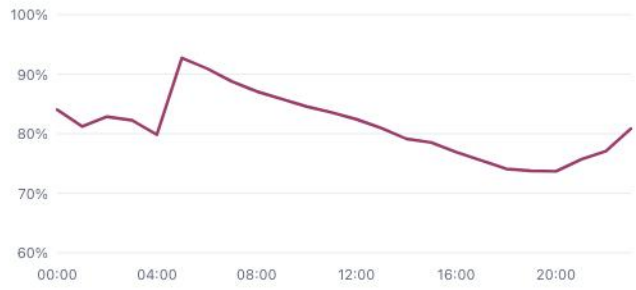
When, where and why flights are late

Descriptive results from the data warehouse: time patterns, the weight of weather, how a late aircraft carries its delay forward, and patterns among airports and cancellations. Delayed means arriving 15 or more minutes late.

Time and airlines | Weather and causes | Delay propagation | Airports and cancellations

On-time share by scheduled departure hour

% of flights, all 2015. Delays build through the day.



By month

February and June are the worst months.

On time % | Cancelled % | Avg delay



By day of week

Saturday is the best day; Monday cancels the most.



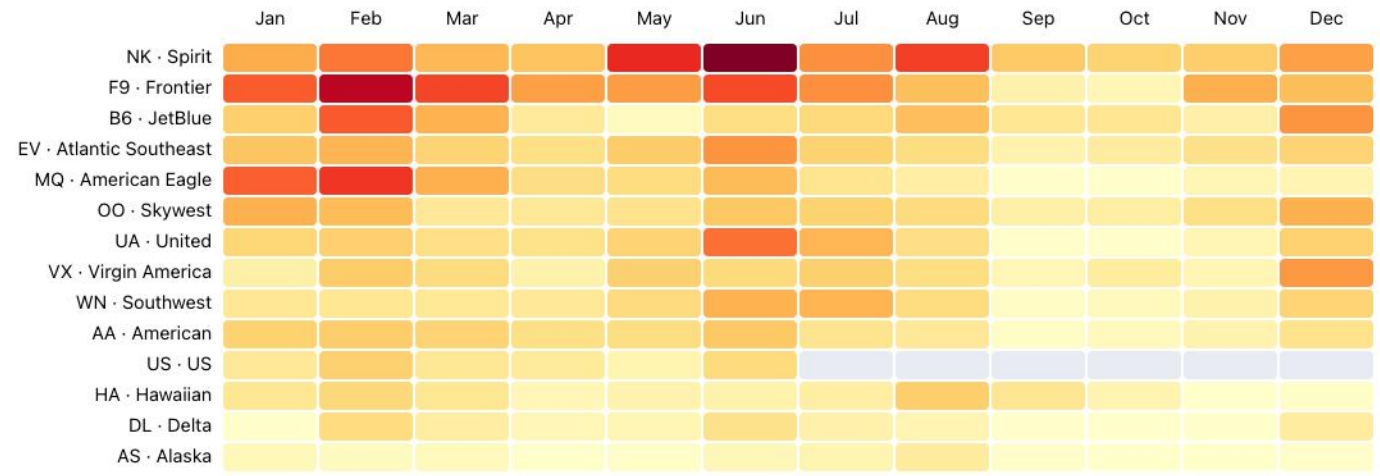
Longest and shortest departure delays

Average departure delay, minutes.



Average arrival delay by airline and month

Minutes; darker is later.



Airlines ranked by on-time share

US Airways covers January–June only. Download CSV

Code	Airline	Flights	On time ▾	Avg delay (min)	Cancelled
HA	Hawaiian Airlines Inc.	76,272	88.7%	2.0	0.22%
AS	Alaska Airlines Inc.	172,521	87.0%	−1.0	0.39%
DL	Delta Air Lines Inc.	875,881	86.4%	0.2	0.44%
AA	American Airlines Inc.	725,984	81.7%	3.5	1.50%
OO	Skywest Airlines Inc.	588,353	81.3%	5.8	1.69%
US	US Airways Inc.	198,715	81.2%	3.7	2.05%
WN	Southwest Airlines Co.	1,261,855	81.0%	4.4	1.27%
VX	Virgin America	61,903	80.8%	4.7	0.86%
EV	Atlantic Southeast Airlines	571,977	80.3%	6.6	2.66%
UA	United Air Lines Inc.	515,723	79.4%	5.4	1.27%
MQ	American Eagle Airlines Inc.	294,632	78.3%	6.5	5.10%
B6	JetBlue Airways	267,048	77.4%	6.7	1.60%

EXPLORE THE DATA

When, where and why flights are late

Descriptive results from the data warehouse: time patterns, the weight of weather, how a late aircraft carries its delay forward, and patterns among airports and cancellations. Delayed means arriving 15 or more minutes late.

Delayed-flight rate by origin weather

% of flown flights. Weather class comes from observed daily precipitation, snow and wind.

Clear weather	17.0%
Moderate weather	27.6%
Severe weather	40.4%

Cancellation rate by origin weather

Clear weather	1.1%
Moderate weather	3.1%
Severe weather	9.0%

Why delayed flights were late

Share of delay minutes, flights 15+ min late.

Late aircraft	39.8%
Airline	32.2%
Air system (NAS)	22.9%
Weather	5.0%
Security	0.1%

> DOT's weather code captures only part of weather's effect: air-system and late-aircraft delays rise on severe days too.

Why flights were cancelled

Share of all cancellations.

Airline	28.1%
Weather	54.4%
NAS	17.5%
Security	0.0%

Weather's share of delay minutes by month

%



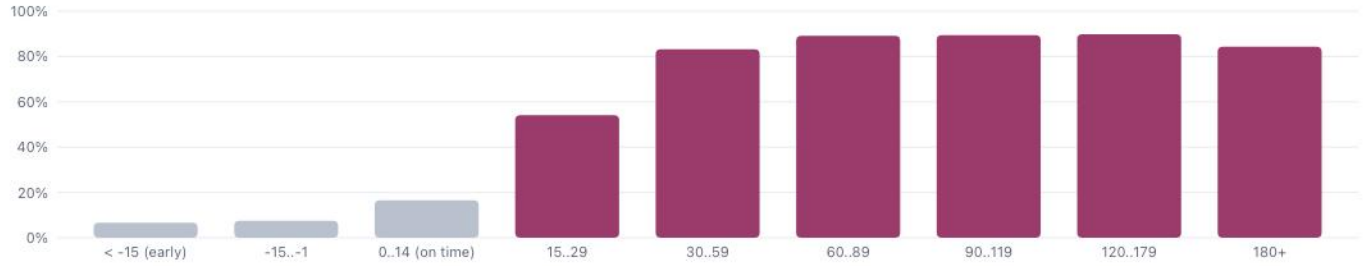
EXPLORE THE DATA

When, where and why flights are late

Descriptive results from the data warehouse: time patterns, the weight of weather, how a late aircraft carries its delay forward, and patterns among airports and cancellations. Delayed means arriving 15 or more minutes late.

Time and airlines | Weather and causes | Delay propagation | Airports and cancellations

Share of departures 15+ min late, by inbound arrival delay Each flight is matched to the same aircraft's previous leg that day.



How much delay a planned turnaround absorbs % of the inbound delay absorbed; negative means the delay grew. Longer turns absorb more.

Inbound delay \ planned turn	<30 min	30-44 min	45-59 min	60-89 min	90-119 min	120-179 min	180+ min
15-29 min	-42%	-15%	5%	50%	73%	74%	32%
30-59 min	-21%	-9%	3%	37%	77%	82%	64%
60-119 min	-12%	-5%	0%	19%	58%	84%	81%
120+ min	-1%	-2%	0%	9%	29%	57%	86%

EXPLORE THE DATA

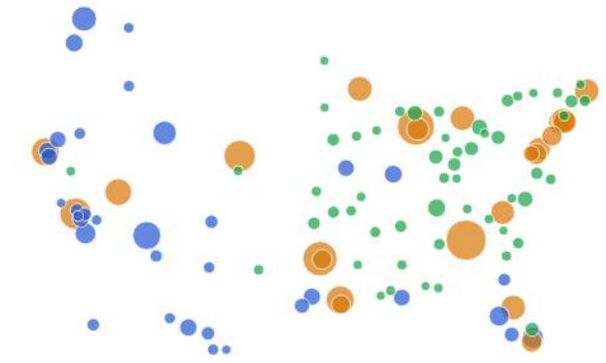
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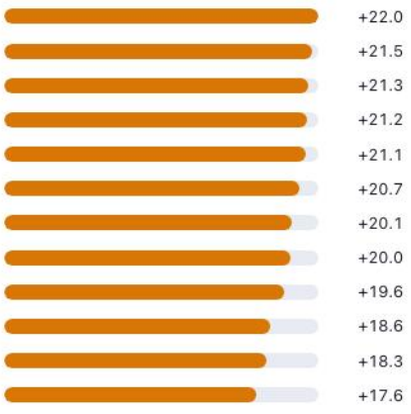
Airports on the map

Colour is the cluster, size is traffic.



Most weather-sensitive airports

Extra delayed flights in adverse vs clear weather, percentage points (airports with 20,000+ flights).



Mid-size, punctual — 35 airports

Delayed flights	**15.9%**
Cancelled	**0.94%**
Weather gap, delay rate	**7.6 pp**
Taxi-out	**12.7 min**
Adverse-weather flights	**10.3%**

PHX, SEA, SLC, FLL, SAN, TPA, PDX, STL

Large hubs, congested — 24 airports

Delayed flights	**21.0%**
Cancelled	**1.77%**
Weather gap, delay rate	**16.9 pp**
Taxi-out	**17.8 min**
Adverse-weather flights	**17.4%**

ATL, ORD, DFW, DEN, LAX, SFO, IAH, LAS

Smaller, weather-exposed — 54 airports

Delayed flights	**18.1%**
Cancelled	**2.33%**
Weather gap, delay rate	**10.7 pp**
Taxi-out	**14.2 min**
Adverse-weather flights	**20.2%**

BNA, CLE, RDU, MKE, IND, CMH, PIT, PBI

Rules most strongly tied to cancellation

Lift = how many times likelier a flight is cancelled than average. Associations, not causes.

Download CSV

If the flight has…	Flights	Cancelled	Share cancelled	Lift ▾	Main reason
airline=MQ & o_snow & d_severe_wx	2,795	1,274	45.6%	27.7×	weather
airline=MQ & o_severe_wx & d_severe_wx	2,576	1,146	44.5%	27.1×	weather
o_severe_wx & o_high_wind	3,542	1,568	44.3%	26.9×	weather
o_snow & o_high_wind	3,359	1,469	43.7%	26.6×	weather
month=February & airline=MQ & o_severe_wx	2,825	1,135	40.2%	24.4×	weather
month=February & airline=MQ & d_severe_wx	2,816	1,100	39.1%	23.8×	weather
airline=MQ & o_severe_wx & o_freezing	4,072	1,565	38.4%	23.4×	weather
airline=MQ & o_freezing & d_severe_wx	3,063	1,173	38.3%	23.3×	weather
airline=MQ & o_severe_wx & o_snow	6,762	2,503	37.0%	22.5×	weather
month=February & weekend & o_high_wind	2,331	828	35.5%	21.6×	weather
o_high_wind & d_severe_wx	2,016	713	35.4%	21.5×	weather
month=February & o_severe_wx & d_severe_wx	4,464	1,559	34.9%	21.2×	weather

ABOUT

Problem, data, method and limits

Problem and relevance

- **Question.** How much does weather change the chance a flight arrives 15+ minutes late or is cancelled, and can we predict it early?
- **Why it matters.** Delays cost passengers and airlines time and money; early warnings let airlines re-time crews and aircraft and airports prepare.
- **Separating causes.** Weather, airline operations, congestion and late aircraft are usually blamed together; we measure them apart.
- **Base paper.** Schösser and Schönberger (2022) on the same data, extended with a stricter time-based evaluation and more models.

Data

- **Flights.** 2015 US DOT on-time performance, 5,819,079 scheduled domestic flights.
- **Weather.** NOAA GHCN-Daily at the nearest station to each airport (precipitation, snow, wind, temperature).
- **Split.** Train January–August, tune on September, test on November–December.
- **October.** Excluded from airport, weather and model work because its airport codes could not be mapped.

Method

- **Warehouse.** Star schema in DuckDB with a verified backup.
- **Models.** LightGBM, XGBoost, Random Forest, Decision Tree, Logistic Regression, KNN, Naive Bayes, an embedding neural network and an ensemble.
- **Rigour.** Class weighting, validation-tuned thresholds, multiple seeds, bootstrap intervals, a leakage check.
- **Explainability.** SHAP, gain importance, ablations and subgroup performance.

Technology

- **Analysis.** Python, SQL on DuckDB, pandas, scikit-learn, LightGBM, XGBoost, PyTorch, SHAP, statsmodels, mlxtend.
- **This portal.** Plain HTML, CSS and JavaScript with d3 for charts; a Python standard-library server with a small JSON API.
- **Predictor.** The saved T-60 LightGBM model (417 trees) with SHAP contributions; the warehouse is opened read-only.

Limits to keep in mind

- **Weather is same-day observed weather** used as a stand-in for a forecast, so its gain is an upper bound.
- **One test period.** November–December only; other seasons are untested.
- **Inbound aircraft.** About three quarters of flights have the inbound aircraft still airborne at T-60.
- **Associations.** Cancellation rules and clusters are patterns, not causes; clusters are soft.
- **Paper comparison is approximate:** different splits, airport subsets and class balancing, and the day-ahead task favours the paper.
- **Predictor.** Rebuilds hidden inputs (exact weather readings, historical delay rates) from typical 2015 values for the route and month.